Q1. (a) What happens if a bar magnet is cut into two pieces: (i) transverse to its length, (ii) along its length? (b) A magnetised needle in a uniform magnetic field experiences a torque but no net force. An iron nail near a bar magnet, however, experiences a force of attraction in addition to a torque. Why? (c) Must every magnetic configuration have a north pole and a south pole? What about the field due to a toroid? (d) Two identical looking iron bars A and B are given, one of which is definitely known to be magnetised. (We do not know which one.) How would one ascertain whether or not both are magnetised? If only one is magnetised, how does one ascertain which one? [Use nothing else but the bars A and B.]

Q2. Figure 5.4 shows a small magnetised needle P placed at a point O. The arrow shows the direction of its magnetic moment. The other arrows show different positions (and orientations of the magnetic moment) of another identical magnetised needle Q. (a) In which configuration the system is not in equilibrium? (b) In which configuration is the system in (i) stable, and (ii) unstable equilibrium? (c) Which configuration corresponds to the lowest potential energy among all the configurations shown?

Q3. (a) Magnetic field lines show the direction (at every point) along which a small magnetised needle aligns (at the point). Do the magnetic field lines also represent the lines of force on a moving charged particle at every point? (b) If magnetic monopoles existed, how would the Gauss's law of magnetism be modified? (c) Does a bar magnet exert a torque on itself due to its own field? Does one element of a current-carrying wire exert a force on another element of the same wire?

Q4. A solenoid has a core of a material with relative permeability 400. The windings of the solenoid are insulated from the core and carry a current of 2A. If the number of turns is 1000 per metre, calculate (a) H, (b) M, (c) B and (d) the magnetizing current I(m)

Q5. A short bar magnet placed with its axis at 30° with a uniform external magnetic field of 0.25 T experiences a torque of magnitude equal to 4.5 × 10 raise to the power 2 J. What is the magnitude of magnetic moment of the magnet?

Q6. A short bar magnet of magnetic moment m = 0.32 JT raiseto the power -1 is placed in a uniform magnetic field of 0.15 T. If the bar is free to rotate in the plane of the field, which orientation would correspond to its (a) stable, and (b) unstable equilibrium? What is the potential energy of the magnet in each case?

Q7. A closely wound solenoid of 800 turns and area of cross section 2.5 × 10raise to the power -4 m2(m square) carries a current of 3.0 A. Explain the sense in which the solenoid acts like a bar magnet. What is its associated magnetic moment?

Q8. If the solenoid in Q9. is free to turn about the vertical direction and a uniform horizontal magnetic field of 0.25 T is applied, what is the magnitude of torque on the solenoid when its axis makes an angle of 30° with the direction of applied field?

Q9. A bar magnet of magnetic moment 1.5 J T raise to the power -1 lies aligned with the direction of a uniform magnetic field of 0.22 T. (a) What is the amount of work required by an external torque to

turn the magnet so as to align its magnetic moment: (i) normal to the field direction, (ii) opposite to the field direction? (b) What is the torque on the magnet in cases (i) and (ii)?

Q10. A closely wound solenoid of 2000 turns and area of cross-section 1.6×10^{-4} m^2, carrying a current of 4.0 A, is suspended through its centre allowing it to turn in a horizontal plane. (a) What is the magnetic moment associated with the solenoid? (b) What is the force and torque on the solenoid if a uniform horizontal magnetic field of 7.5×10^{-2} T is set up at an angle of 30° with the axis of the solenoid?

Q11. A short bar magnet has a magnetic moment of 0.48 J T^{-1}. Give the direction and magnitude of the magnetic field produced by the magnet at a distance of 10 cm from the centre of the magnet on (a) the axis, (b) the equatorial lines (normal bisector) of the magnet.

Q12. A short bar magnet placed in a horizontal plane has its axis aligned along the magnetic north-south direction. Null points are found on the axis of the magnet at 14 cm from the centre of the magnet. The earth's magnetic field at the place is 0.36 G and the angle of dip is zero. What is the total magnetic field on the normal bisector of the magnet at the same distance as the null–point (i.e., 14 cm) from the centre of the magnet? (At null points, field due to a magnet is equal and opposite to the horizontal component of earth's magnetic field.)

Q13. If the bar magnet in exercise 5.13 is turned around by 180°, where will the new null points be located?

Q1: Cutting a Magnet & Magnetic Properties

Concepts: Magnetic poles, dipole moment, pole strength

1. A bar magnet of length **15 cm** is cut into three equal parts along its length. How does the magnetic moment of each part compare to the original?
2. A **U-shaped magnet** is cut into two halves symmetrically. Sketch and explain the magnetic poles of each piece.
3. If a bar magnet is **cut at an angle of 45°**, how will the pole strength and dipole moment change?
4. A magnetized **sphere** is placed inside a uniform field. Will it exhibit distinct poles? Why or why not?
5. A rectangular magnet is split into **two unequal pieces**. Explain how the **distribution of magnetic poles** changes.

Q2: Magnetic Equilibrium of Dipoles

Concepts: Torque, stability, equilibrium positions

6. A magnetic dipole is placed in a **non-uniform field**. Will it experience both force and torque? Explain.
7. Two identical bar magnets are suspended in Earth's field at **different angles**. Which one will oscillate faster? Why?
8. If a **third identical magnet** is placed **perpendicular** to an existing bar magnet, describe the stability of the system.
9. A bar magnet experiences **a torque but no net force** in a uniform field. Justify mathematically.
10. A **floating magnet system** is disturbed slightly from equilibrium. What kind of motion will it exhibit?

Q3: Magnetic Field Lines & Forces

Concepts: Field lines, charged particles, Lorentz force

11. Why do magnetic field lines **never intersect**? Give a mathematical explanation.
12. A **proton** moving at **2×10^6 m/s** enters a **0.1 T field** perpendicular to its motion. Calculate its **radius of curvature**.

13. How does a **charged particle** behave in **parallel vs perpendicular fields**? Use equations.
14. If the **Earth's field suddenly vanishes**, how would a compass behave?
15. Draw the field lines for a **bar magnet surrounded by a soft iron ring**.

Q4: Magnetic Field & Magnetization in a Solenoid

Concepts: Magnetic intensity, permeability, Ampere's Law

16. A solenoid with **2500 turns/m** carries **3 A** current. If the core has a permeability of **600**, find **H**, **B**, and **M** inside it.
17. A solenoid is wound on a **cylindrical iron core**. How will increasing the core's **permeability** affect its field?
18. Compare the **field at the center** vs **ends of a solenoid**. Why is it different?
19. Explain how the **magnetic field inside a toroid** differs from that inside a solenoid.
20. A solenoid carries a **time-varying current**. What happens to its internal field?

Q5: Torque on a Magnetic Dipole

Concepts: Work, torque, dipole moment

21. A bar magnet of **moment 0.8 $A \cdot m^2$** is placed at **45°** in a **0.3 T** field. Find the **torque** acting on it.
22. If the **external field is tripled**, how does the **torque on a dipole** change?
23. A **dipole initially at 0°** is rotated to **90°** in a field of **0.4 T**. Find the **work done**.
24. Can a dipole experience **zero torque** but **nonzero potential energy**? Explain.
25. A **circular loop of current** behaves like a dipole. Derive an expression for its **torque in a field**.

Q6: Magnetic Moment and Equilibrium of a Bar Magnet

Concepts: Stability, oscillations, energy

26. Why is **0° a stable** and **180° an unstable** equilibrium position for a magnetic dipole in a uniform field?
27. Derive an expression for the **time period of oscillation** of a freely suspended bar magnet.

28. Two magnets with **different dipole moments** are suspended in a field. Which one will oscillate faster?
29. A **bar magnet is heated** to high temperatures. What happens to its magnetic moment?
30. How does the **mass of a magnet** affect its time period of oscillation?

Q7: Solenoids & Magnetic Moment

Concepts: Field equations, dipole analogy

31. A solenoid behaves like a bar magnet. Justify using the **Biot-Savart law**.
32. A solenoid with **500 turns per meter** carries **4 A**. Find its **magnetic moment**.
33. Derive an expression for the **energy stored in a solenoid's field**.
34. If a **solenoid is bent into a ring**, how will its field distribution change?
35. Why is a solenoid's **field stronger than that of a bar magnet**?

Q8: Torque on a Solenoid in a Magnetic Field

Concepts: Interaction with external fields

36. A solenoid in a **0.3 T field** is tilted by **60°**. Find the **torque** acting on it.
37. If the **number of turns** in a solenoid is doubled, how does its **torque change**?
38. Explain why a **solenoid in a uniform field does not experience a net force**.
39. Show that a solenoid's **energy in an external field** is similar to that of a magnetic dipole.
40. Why does a **current-carrying solenoid** align with Earth's field?

Q9: Work Done on a Bar Magnet in a Magnetic Field

Concepts: Potential energy, alignment

41. A magnet of **1.5 A·m^2** is rotated from **parallel to perpendicular** in a **0.5 T field**. Find the **work done**.
42. Derive the equation for **potential energy of a dipole** in a field.

43. If a bar magnet is rotated continuously in a field, how does its **energy change over time**?
44. A magnet is held at **60°** in a **uniform field**. Find the **work required to rotate it to 120°**.
45. Explain why a **bar magnet prefers aligning** with an external field.

Q10-Q13: Advanced Mixed Problems

Concepts: Magnetization, null points, experimental methods

46. Why does the **Earth's magnetic field** cause a bar magnet to rotate but not translate?
47. If a **magnet is placed in a stronger field**, does its dipole moment increase?
48. A **current loop is placed in a non-uniform field**. Explain why it experiences a force.
49. A bar magnet's null points are found at **20 cm**. What happens if the magnet is **reversed**?
50. Explain the **role of Earth's core in sustaining the geomagnetic field**.
51. A bar magnet's strength **halves** due to demagnetization. How does this affect its null points?
52. A magnet at a **30° tilt** creates a field of **0.2 T**. What is the field at its equator?
53. If a **magnet is slowly rotated**, how does its **potential energy curve look**?
54. A bar magnet in Earth's field experiences a force. Explain why it is hard to measure.
55. Why do cosmic particles **spiral around Earth's field lines**?